

A Comprehensive Review of Three and Five Level Transformerless Inverter Topologies Based on Component Count and their Performance

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Abstract

Abstract: In solar photovoltaic (SPV) applications, transformerless grid-tied inverters have garnered significant research interest because of several promising characteristics such as higher efficiency, tolerable power density, lightweight design and affordability. Among the different types of grid-tied SPV inverters, transformerless multilevel inverters (MLI) stand out as particularly popular. This popularity stems from their capacity to transmit reactive power effectively, their common-ground (CG) configuration, which helps in mitigating the detrimental effects of leakage current caused by the parasitic capacitances of the PV array and their ability to accommodate smaller size filters for minimizing total harmonic distortion (THD). Several transformerless inverter (TLI) concepts have emerged in recent times to effectively eliminate leakage current. These approaches employ various strategies that include decoupling the DC from the AC side, implementing clamping mechanisms for controlling common mode voltage (CMV) during the freewheeling phase, and adopting CG configurations. Over the years, numerous new topologies and configurations came into existence, and it becomes a complex and confusing task while selecting the topology for SPV. Hence, to provide a clear overview of the advancements in TLI for grid-tied SPV systems, this paper seeks to classify and thoroughly review different TLI models. The comparison among the different topologies was conducted based on size, leakage current elimination, THD, total stress voltage, and other relevant factors. The findings were then summarized in a table for a concise representation of the comparative analysis. This will help researchers in selecting which topology is best for particular SPV application.

1. Introduction

Power converters play a significant role in integrating a renewable energy (RE) sources with the electric grid. In a grid-linked PV system, the PV input is connected to the local grid with the help of inverters, allowing the flow of AC power into the grid. There are two main categories of grid tied inverters: galvanic-isolated system and non-isolated system. Generally, galvanic isolation is achieved by connecting a high-frequency (HF) transformer to the DC terminal of the inverter or a line frequency (LF) transformer to the AC terminal. The transformers on AC side are bigger in size, more costly, difficult to install. On the other hand DC-side transformers are less expensive and smaller than AC-side transformers. The system becomes more complex with the addition of more power conversion stages. Thus, grid-connected TLI are gaining popularity in RE applications due to their smaller size, lower cost and increased efficiency. However, as shown in the Figure 1, there exists a leakage current in the absence of transformer, because of the presence of parasitic capacitance (C_{PV}) between PV-to-ground. The voltage across C_{PV} is known as the CMV and the relationship between instantaneous leakage current ($I_{Leakage}$) and CMV is as follows:

$$I_{Leakage} = c_{pv} \frac{d(CMV)}{dt} \quad (1)$$

The switching HF and line LF components that make up leakage current introduce harmonics into the grid

current, elevate the losses of system, increases electromagnetic interference (EMI) and may pose a risk to public safety such as electric shock. Thus, reducing the leakage current is important.

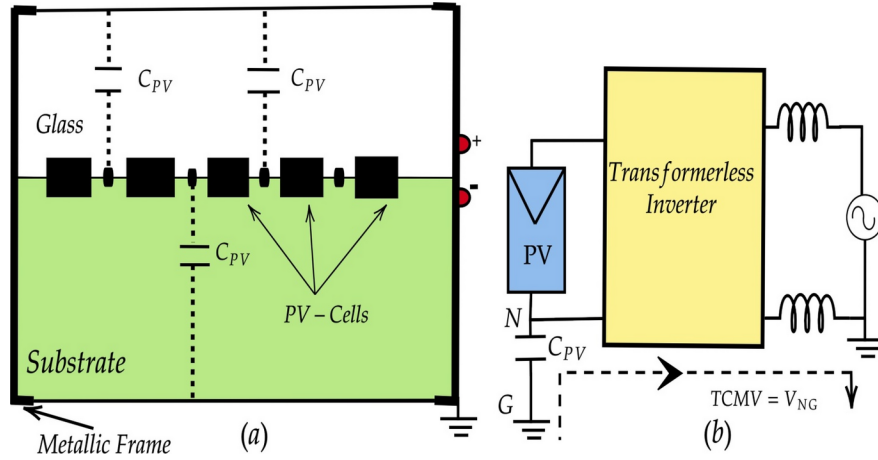


Figure 1: (a) PV panel cross section demonstrating the occurrence of C_{pv} between the PV cells and the metallic frame (b) circuit diagram of a TLI showing the path of leakage current and CMV.

The leakage current issue has been tackled by many researchers employing various techniques, all of which focus on maintaining CMV constant or minimizing or removing the HF ripple component of CMV. In the existing literature, numerous TLI topologies have been suggested with the purpose of mitigating the leakage current. Thus, many decoupling techniques have been used to accomplish it, including clamping the CMV during the freewheeling phase, separating DC from the AC side or using CG configuration. These techniques combined with integrated MPPT circuits generate numerous topologies that are often difficult and confusing to follow. Hence, to provide a comprehensive overview of TLI development for the grid-tied SPV system, this paper seeks to:

- Reviewed and categorized single-phase three and five level TLI topologies.
- Presented a detailed analysis of state of art single-phase three and five level TLI topologies.
- Compared these TLI topologies based on component count, THD, leakage current, TSV and efficiency analysis.
- Summarized the presented topologies which will help researchers in selecting among these topologies which TLI topology is best for a particular application.

This classification and analysis can assist researchers in understanding the advantages and disadvantages of different TLI topologies in terms of their leakage current and CMV behavior. The rest of the paper is divided into the following sections: a general classification of various TLI topologies is provided in section II. In Section III, a tabular comparison among different TLIs has been drawn and Section IV discusses the future recommendations and author inference. Finally, Section V provides the conclusion.

2. Classification of Transformerless Inverter Topologies

In TLI there is not any galvanic isolation between the source and load. The lack of galvanic isolation results in leakage current flow that exists caused by the parasitic capacitance between PV panel to ground. In this context, numerous TLI topologies have been examined in the literature. All these topologies use one of the following method to lower the leakage current:

1. Maintaining a constant CMV by isolating PV array from the grid
2. Connecting negative terminal of PV array with the grid neutral, also known as CG type
3. Connecting the grid neutral to the middle of the DC-link bus

Figure 2 illustrates the categorization of single-phase TLI topologies, where the single-input group is classified into three types: common ground, buck-boost type (BBT) and H-bridge type topologies. H-Bridge type are classified into two groups i.e. Decoupling and Mid-point clamping. Decoupling is further classified into two sub-groups i.e. AC-Decoupling and DC-Decoupling.

The classification is based on keeping the CMV constant or having it fluctuate at a lower frequency. This is accomplished by isolating the DC or AC terminals of a conventional H-bridge inverter, such as H-5, H-6 etc. during the zero level mode. In H-Bridge based inverter topologies, the parasitic capacitance of switches makes it extremely difficult to limit the ground leakage current. Furthermore, these inverters need two filter inductors, which rises their cost and size. Additionally, the conduction losses during active mode are high because more than three switches are connected in series.

Another option is to use half-bridge topologies, in which grid neutral is connected to the midpoint of the DC-link. This type of inverter uses one filter inductor and two switches to mitigate the leakage current. These topologies result in lower efficiency and higher costs since they need additional boosting circuits to meet grid requirements.

Another category is CG based inverter topology, where the parasitic capacitance between the PV array and the ground is short-circuited as neutral of grid is directly connected to negative terminal of PV array. CG-based inverters totally eliminate leakage current, in contrast to techniques that seek to maintain CMV constant and hence minimize the leakage current.

In the following section, CG based inverter topologies are reviewed in detail and presented in a table for comparison.

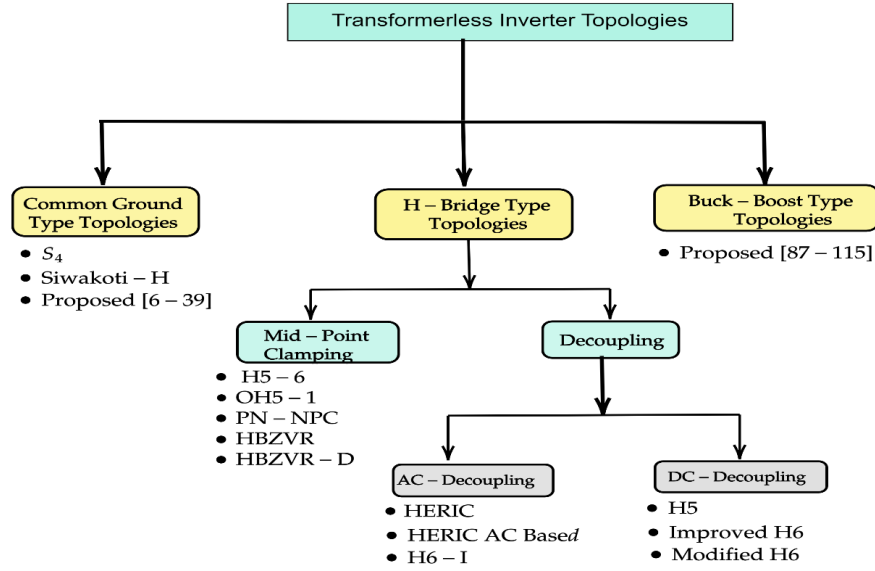


Figure 2: Classification of single phase TLI topologies.

3. Common Ground Topologies

This section discusses the CG based inverter in detail. The configurations where the neutral of grid is linked directly to negative terminal of PV panel are termed as CG type topology, like S4 [1], Siwakoti-H [2], and those presented [3], [4] and [5], etc. A notable benefit of these topologies lies in the constant CMV and

the eradication of leakage current. In this setup, the parasitic capacitance is effectively mitigated through a direct connection made between the PV panel neutral and grid neutral.

The S4 topology presented in [1], employs sinusoidal Pulse Width Modulation (SPWM) for the operation of inverter. The utilization of SPWM serves to minimize switching losses and reduce the need for extensive filtering. The topology put forth in [2], demonstrates a substantial decrease in the number of semiconductor components, utilizing merely four switches. Configured akin to an H-bridge, this inverter generates a negative voltage specifically during the negative cycle of inverter operation using a flying capacitor (FC). The FC concept finds application in CG-TLI topologies, the initial topology was introduced by Siwakoti and Blaabjerg in paper [2], with the subsequent one proposed by them in paper [3]. Both topologies operate with identical modulation pulses. The topology introduced in [4], presented a virtual DC bus concept and the primary objective of this approach was to produce the required negative voltage at the output essential for functioning as an inverter. Notably, the topology incorporates a CG connection. Hence, zero leakage current is achieved as the parasitic capacitors (C_{pv1} and C_{pv2}) are held at the neutral potential. Figure 3 displays the circuit configuration.

The paper [5], introduced a CG single-phase TLI inverter, it has four switches, a capacitor, a diode and a filter at the output. It is modulated using the standard SPWM approach. In contrast to conventional topologies, all components in this topology are subjected to a voltage stress of equal magnitude that is equal to the voltage across the DC link. It is possible to regulate the flow of active and reactive power to the grid.

The paper [6], presented a 5-level single-phase TLI topology featuring FC and switched diodes. The suggested configuration makes use of reduced number of switches and incorporates self-voltage-balancing for the FC, voltage boosting capability, and effectively suppresses leakage currents. An appropriate modulation strategy is used in the proposed TLI to produce CMV with fewer voltage swings and transitions. The paper suggested a modification to the IH6 topology, retaining all its advantages. The proposed modification doubles the peak output voltage for the same input DC voltage while enhancing power quality. The inherent boosting ability and generation of high-quality voltage by this modification abolish the need of additional circuitry and bulky filtering components. The inverter topology is illustrated in Figure 4 and comprises of seven fully controlled switches.

The topology proposed in [7], is a 5-level transformerless boost inverter. It is illustrated in Figure 5. It employs nine switches and two FC to generate five distinct output voltage levels: $-2V_{DC}$, $-V_{DC}$, 0, V_{DC} , and $2V_{DC}$. To achieve this, S_1 must be capable of bidirectional voltage blocking, which can be realized through a series connection or by using a device equipped with a reverse blocking diode, as illustrated in Figure 5. Depending on the desired output voltage, the FC are charged in parallel and discharged either in parallel or series, during the inverter operation. Notably, the topology incorporates a CG connection, hence leakage current is effectively eliminated, thus contributing to improved performance.

The suggested topology in [8], is comprised of six switches, two capacitors, two diodes and an inductor (Figure 6). The switches used are unidirectional with antiparallel diodes. The system operation relies on the series-parallel switching of capacitors, effectively making them function as a virtual DC link. This approach is analogous to techniques employed in conventional MLI. The proposed topology achieves a single-stage boost of the input PV voltage through the controlled charging and discharging of the capacitors. The suggested topology [9], incorporates CG connection of inverter (Figure 7), to lower the leakage current. Additionally, the suggested converter incorporates a FC leg along with an H-Bridge (HB) cell, effectively synthesizing a multilevel voltage with a lower harmonic content. This configuration attempts to elevate the overall performance and efficiency of the inverter system. Furthermore, the proposed inverter has the capability to handle reactive power and also it exhibits a maximum utilization of DC bus voltage, which stands in contrast to HB-based topologies.

The paper [10], introduced a modified H-bridge-based MLI topology. The configuration is derived from a conventional single-phase H-bridge, with one leg modified and is connected to the middle of a split DC-link through a bidirectional switch known as a T-type leg. Meanwhile, the other leg remains unmodified,

representing a usual 2-level leg. Thus, it is termed as asymmetrical T-type 5-level transformerless (5L-T-AHB) single-phase topology. A key feature of this topology is that the two-level leg operates at the 50 Hz frequency. It is shown in Figure 8. The paper [11], puts forward a novel 5L-CG-TLI incorporating a SC unit, ensuring 100% utilization of DC-bus for SPV system as depicted in Figure 9. The suggested topology incorporates a CG connection. Thereby, effectively eradicating CMV. A simple series-parallel technique is employed to balance the voltage across the SC. Additionally, the inverter control is facilitated by a simple PWM scheme. In contrast to HB topology, the suggested topology is capable of self-voltage balancing of SC and full utilization of DC-bus.

The paper [12], introduced a novel CG-5L-TLI. The suggested topology consists of a switched-boost (SB) module connected to a dual T-type (D2T) cell, having two bidirectional and six unidirectional an inductor and two DC-link capacitors. By adjusting the duty cycle of the SB module in conjunction with one switch from the D2T cell at each moment, the topology achieves a valuable voltage boosting capability, enabling it to adapt to wide variation in input voltage. The topology named SBD2T5L-TLI not only offers voltage boosting capabilities but also includes a soft charging operation ability, making it a suitable topology for SPV systems. It is shown in Figure 10.

The paper [13], presented an enhanced cascaded MLI designed for highly efficient and reliable performance, particularly focused on minimizing leakage current. The proposed scheme not only reduces the switch count, but also exhibits low conduction and switching losses. The topology, combined with a specific PWM technique, effectively diminishes high-frequency voltage transition in CMV. This reduction in high-frequency voltage transitions contributes to minimizing leakage current and reducing the size of EMI filters. The topology uses eight switches to produce five levels of output voltage. The four switches in this topology are operated at low SF to reduce the switching losses. The inverter can be cascaded easily to obtain more than five levels in the output voltage. The THD of the grid current complies with the requirements of the IEEE 1547 standard. This configuration offers improved performance, reduced harmonic distortion, and compliance with industry standards. The circuit diagram of the suggested topology is depicted in Figure 11. The suggested topology consists of two converters (Conv-I and Conv-II). Conv-I is a HB inverter consisting of two switches S_{x1} and S_{x2} . The Conv- II consists of six switches (S_{x3} - S_{x8}) that constitute an H-bridge.

The paper [14], introduced a family of single-phase SC-5L-TLI topologies for SPV application. The proposed family includes four distinct inverter configurations like H8, H10, H8-UPF and H10-UPF. These configurations are designed to mitigate leakage current by keeping a constant grid frequency voltage at the PV parasitic capacitor, even while considering switch terminal capacitances. An important characteristic of the SC-5L-TLI family is the achievement of 100% utilization of DC-bus, constant CMV and symmetrical operation. The general structure of the suggested SC-5L-TLI family is depicted in Figure 12, comprising three main components: a SC block, four-quadrant switch (4QS) and a conventional H-bridge. The SC block can be configured in two possible setups: SCB-I and SCB-II, are depicted in Fig. 12(b) and 12(c), respectively. Any SC blocks can be chosen based on the requirement of reactive power capability. The different members of this family are obtained through different combinations of SCB-I and SCB-II and 4QS.

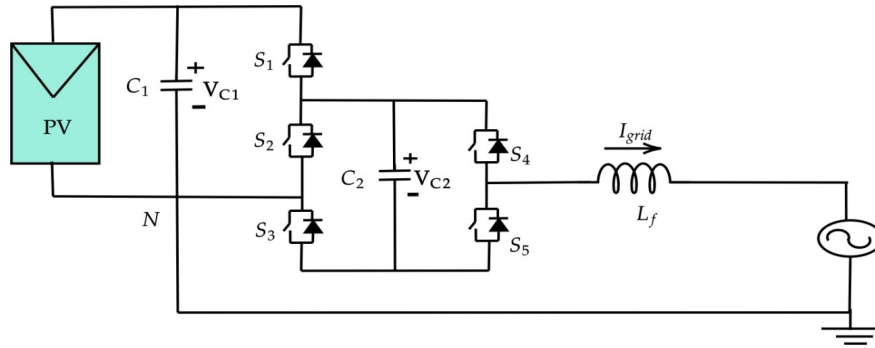


Figure 3: Virtual DC bus based CG- TLI.

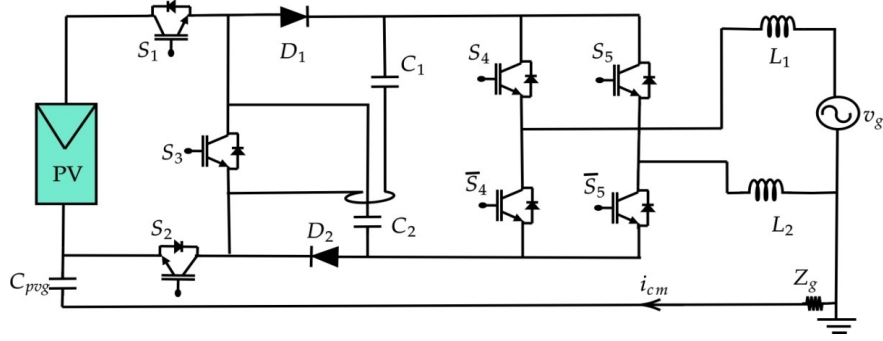


Figure 4: Single-phase five-level TLI topology.

Figure 5: Transformerless five-level boost inverter.

The paper [15], introduced a novel CG-5L-TLI designed to enhance the performance of grid-connected system through its inbuilt boosting capability and unipolar PWM scheme. By integrating a SC module and employing the concept of virtual DC-link, double voltage boost can be achieved within a single-stage. The suggested TLI is illustrated in Figure 13. This setup utilizes six switches, a diode and two capacitor. In this topology, capacitor C_2 functions as a virtual DC-link, analogous to conventional approach. The suggested topology is based on CG concept. This design choice helps in minimizing CMV and leakage currents, enhancing overall system efficiency and performance. Here, four switches are unidirectional and equipped with antiparallel diodes, allowing current flow in a single direction and facilitating freewheeling when switches are turned off. Switch S_3 is also unidirectional but lack an antiparallel diode. Furthermore, switch S_5 is bidirectional, capable of conducting current in both directions and blocking PIV in both polarities, thus enhancing the circuit's resilience and efficiency.

Figure 6: Five-level grid-tied TLI.

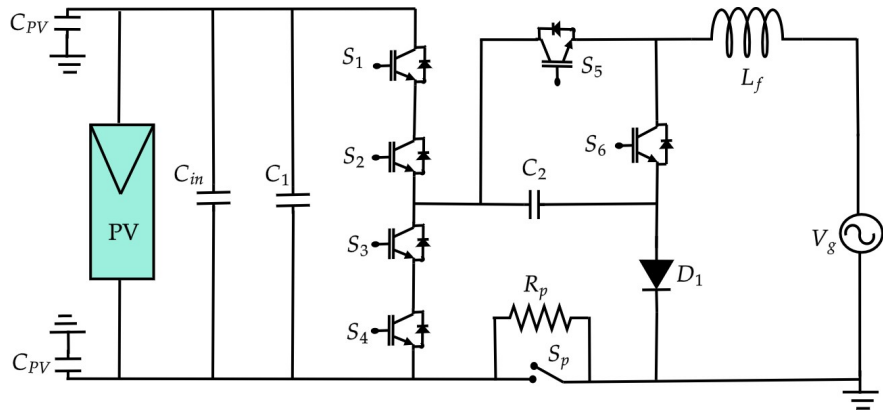


Figure 7: TLI for single-phase SPV application.

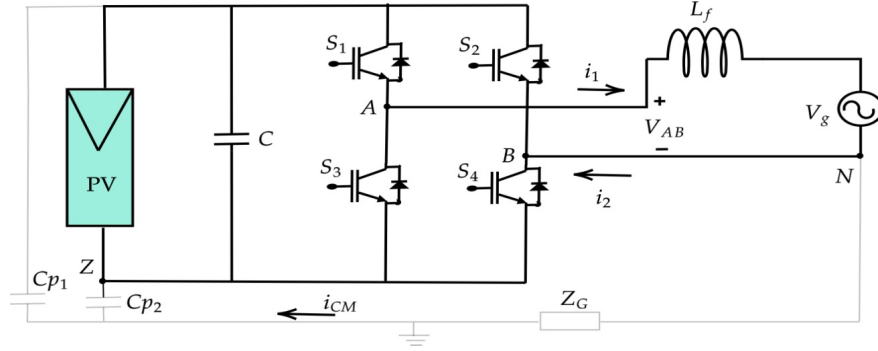


Figure 8: Asymmetrical T-type five-level TLI.

The paper [16], presented a novel type of 5L-CG-TLI designed to elevate the performance of grid-connected system. The presented topology is built upon the SC cell, capable of producing output voltage with five levels using six unidirectional switches. In a single-stage operation, the inverter obtains a double voltage boosting feature by employing series-parallel switching conversion of the SC cell. As a result, a significantly smaller value of input voltage can be employed to attain the peak voltage requirement of the grid. The general setup of the proposed 5L-CG-TLI is illustrated in Figure 14. It incorporates an SC circuit operating via series-parallel switching conversion. This SC circuit consists of a DC source, a capacitor, a power diode and two HF series-parallel switches (S_p and S_s). Additionally, the SC cell is interconnected with four other switches and an additional diode/capacitor for connection to the grid, facilitated by L- filter.

The paper [17], introduced a simplified SC inverter, which utilizes a combination of a Marx inverter structure and an H-bridge. By toggling capacitor between series and parallel configuration, the output voltage of this inverter can be greater than the input voltage. Unlike traditional SC inverters, this design omits the need for bulky inductors, contributing to a more compact system. Additionally, the multilevel output of this inverter helps in mitigating output harmonics.

The paper [18], presented a novel SC- MLI topology designed to address both voltage boosting and leakage current elimination. In this proposed design, capacitors serve as a virtual DC power source to amplify the input voltage. By uniformly charging all capacitors from a single DC source, a staircase waveform is achieved, eliminating the need for multiple DC sources. Moreover, to generate negative voltage levels, capacitors are connected to the output in reverse polarity, obviating the traditional H-bridge circuit typically used in SC-MLI. The CG connection of suggested topology enables leakage current elimination. The topologies introduced in the papers [19] and [20] have demonstrated remarkable voltage boosting capabilities. However, these topologies suffers from the drawback of using excessive switches. Additionally, these topologies require the use of four capacitors to achieve five levels of voltage at the output. This proliferation of capacitors not only adds bulkiness to the inverter circuit but also enhances cost of the inverter and elevates the risk of capacitor failure. In the paper [21], another well- balanced topology has been presented which employs six switches, three SC, one diode, and a DC source. This configuration offers active and reactive power supply capabilities and adopts a CG structure with a simple control scheme. However, it utilizes three SC to achieve five-level of output voltage. Furthermore, four switches in the topology operate at HF, resulting in increased switching losses.

The paper [22], suggested a novel step-up 5-Level CG-TLI inverter designed for grid-connected SPV application. The suggested inverter relies on the FC concept, utilizing a DC source, seven switches and two capacitors to generate 5-level output voltage. The single-stage topology has several advantages, including its CG configuration, support for reactive power, voltage boosting ability, and a reduced component count. These features render it a highly desirable structure for grid-tied SPV application.

The paper [23], introduced a novel six switch 5L- CG inverter for SPV system, utilizing a single DC voltage source. The suggested 6S-5L-CG inverter offers key advantages, including inherent mitigation of leakage

current, capability for reactive power transfer, minimized component count compared to existing inverters, uniform voltage stress distribution over power switches, reduced capacitor size, self-balancing of capacitor voltages and higher efficiency by activating only two switches at a time during each operational state.

The proposed inverter in [24], utilizes four switches and three diodes, all rated at the input voltage V_{dc} . Additionally, the topology incorporates two capacitors, C_{in} and C_1 , along with a magnetically coupled inductor. Furthermore, the suggested inverter features only four switches, distinguishing it as the first 5L-CG topology in recent literature to utilize fewer than six switches. Additionally, the topology minimizes high-frequency operation, with only two switches being active in this mode. A 5-level inverter presented in [25] and [26] has lower component count, but it needs current and voltage sensors that raises the cost. An inverter topology having zero leakage current and a significantly decreased component count was proposed in paper [27]. Two SC are employed to enable the boosting feature of the suggested topology and there is no restriction on modulation index because SC are balanced naturally. The paper [28], presented a novel 5L-TLI topology designed to eliminate leakage current using a SC approach. Additionally, the proposed topology inherently maintains balanced voltage across its capacitors, eliminating the need for a dedicated control strategy to regulate SC voltage. The paper [29], introduced a novel dual-mode 5L-CGT TLI topology capable of accommodating a large input range from 200 V to 400 V. This topology lower the DC-link voltage requirements to as low as one-fourth of conventional topologies, while simultaneously eliminating the leakage current.

The paper [30], introduced a five-level TLI design centered around a single SC, featuring only seven switches, a DC voltage source and a single capacitor. The suggested TLI also includes auto-boosting feature. Importantly, the switching loss within the inverter is reduced as only three switches operate at HF. The paper [31], introduced a new concept for CG type topology that eliminates the need for SC circuit. The newly developed CG inverter topology is a single-stage derived from a combination of front end 3-level boost converter and a classical T-type inverter. The topology mitigates high inrush currents and simultaneously addresses CMV issues while maintaining a continuous DC current.

Figure 9: 5L-CG-TLI.

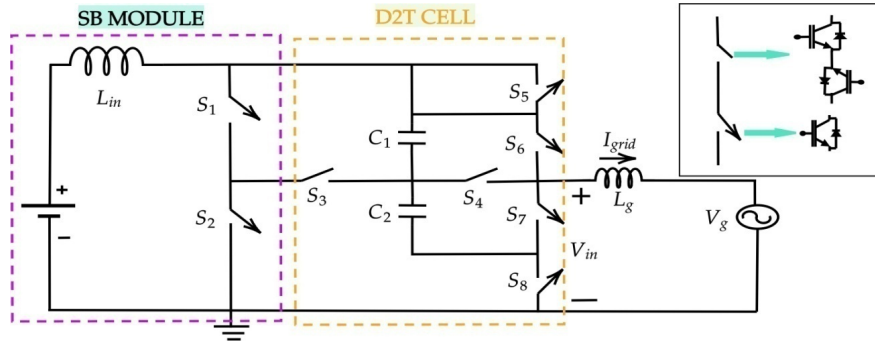


Figure 10: SBD2T5L- TLI.

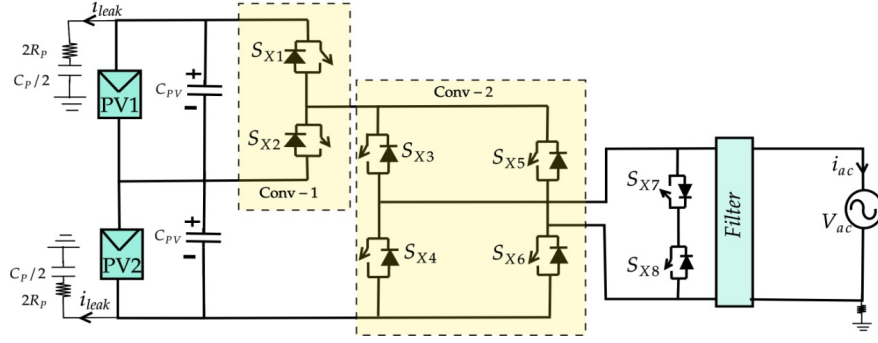


Figure 11: Five- level cascaded MLI for PV system.

The paper [32], introduced a novel 5-level FC inverter designed for grid-connected SPV system. The topology is capable of accommodating increasing input DC voltage and completely eliminating leakage current. It comprises of eight switches, two capacitors and a diode. The inverter structure offers automatic capacitor voltage balancing and also enables the exchange of reactive power, further enhancing its advantages. In paper [33], a fundamental 5L-CGSC-TLI is introduced, utilizing eight switches and two capacitor. In the proposed topology, QSC technique for charging the capacitor is used and a generalized configuration for multiple levels of output voltage is also proposed. The paper [34], presented a new SB-based CG-TLI capable of producing 5- level of output voltage. The inverter is named as SB-CG5L-TL, it offers high voltage gain in a single power conversion stage. By integrating a simple diode-capacitor-inductor (DCL) network into the SB module, a voltage gain of four times is obtained, thus maintaining a distinctive five level output without extra power switches. The article [35], introduced a novel CG-5L-TL -SC based inverter with an inbuilt boost converter. This topology enables soft charging of the SC and reduces count of components. It achieves higher gain by combining inductor-based boosting with SC-based boosting. Unlike existing CG TLI, this design ensures SC charging in every switching cycle for 5 level operation.

The paper [36], introduced a new CG-TLI designed to enhance grid-connected system performance through inbuilt voltage boosting and use of unipolar PWM strategy. The topology eradicates leakage current entirely and offers reactive power control without affecting leakage current. It increases output voltage twice its value without needing an additional boost stage by implementing an SC module.

The paper [37], introduced a new type of CG-TLI. The proposed structure enhances conventional grid-tied system performance due to inbuilt boosting capability and use of unipolar PWM method. A boost factor of two is achieved in a single-stage operation without the need for auxiliary inductor by employing a Virtual DC-link and SC module concept. Furthermore, a PCC strategy is adopted to precisely control the current supplied to the grid.

In paper [38], a novel 5L SC-based CG-TLI topology is introduced, where the concept of the CPC cell, typically utilized in CG-TLI is implemented through an SC cell. This provides voltage boosting which is achieved in a single stage. The production of five level output voltage is facilitated by employing an inductor-based filter at the output. This approach eliminates challenges such as active damping issues and resonant problem related with higher order LCL or LC filters. An adaptive hysteresis current controller technique is employed to achieve the ultimate goal of injecting high-quality AC current into the grid.

The article [39], introduced a new 5-level TLI CG type topology, aiming to reduce voltage stress across circuit elements. The design features fewer components and a simplified structure. The article proposes the use of QSC technique in order to address high charging current during capacitor charging. The paper [40], proposed a 5-level SCMLI for SPV applications. It employs CG configuration to reduce the leakage current, using only seven switches and two capacitors. The efficiency of this topology is 95.03% at a load of 2-kW. The paper [41], presented a 5-level inverter with a CG and inbuilt boosting ability. The CG design eliminates leakage current, making it suitable for SPV applications. A major advantage of this inverter

is that it uses two SC, both rated for only half the output voltage, with self-balancing properties, leading to a smaller, affordable, and simpler design. Furthermore, the topology features lower component count a (seven switches and two diode) lower TSV, low dv/dt . The paper [42] introduced a CG 5-level TLI for solar photovoltaic applications, having lower switch count. The inverter uses four switches, two capacitors, three diodes, and two inductors. Notably, only two switches run at HF, which leads to lower switching losses. The interleaved operation of a coupled inductor enhances the quality of output current by shifting the first set of harmonic components to around twice the SF. Furthermore, it makes full use of the DC bus voltage and has reactive power capabilities. The paper [43] suggested a 5-level TLI using a SC approach. The topology uses seven switches, a capacitor and an inductor that limits high inrush current. The lower component count minimizes the inverter losses. The topology uses LSPWM technique in order to achieve reactive power ability and self-balancing of the SC. A simple closed-loop controller is used for balancing of voltage, providing a stable voltage and reliable grid-connected operation of inverter. The efficiency of the inverter is 97.3%. The paper [44], introduced a single-phase, single-stage SB TLI inverter with CG configuration for SPV application. Since the inverter employs CG configuration, it leads to smaller leakage current, lower power loss, and less harmonics in the inverter. Furthermore, the larger voltage gain of the inverter enables it to function with a reduced shoot-through (ST) duty ratio and a high modulation index, which lowers the DC link voltage need across the inverter leg, resulting in lower voltage stress across the switches. A modified PWM control technique is employed, where the dual active states of the inverter lower current ripple, making it suitable for SPV applications. It is also possible to voltage boosting capability of the inverter by introducing more intermediate impedance sub-networks. The paper [45], suggested a new CG topology for a 5-level TLI tailored for SPV systems. Since it uses CG configuration, the leakage current issue is addressed. The topology uses only five switches, two input capacitors, a diode and a SC to generate the 5-level output voltage. Only two switches are triggered in order to produce positive and negative voltage levels. The charging current of the SC is controlled by a front-end inductor. Additionally, LS-PW technique is used to produce pulses for gate.

In CG topologies, there is a complete elimination of leakage current that makes TLI topologies more suitable for SPV applications. In CG topologies, more effort is put into reducing the switch count, which increases overall efficiency and reduces the system size. The CG topologies are presented in summarized form in Table 1. The paper [6], suggested a modification to the IH6 named topology, retaining all the advantages of the IH6 inverter. The topology presented in [8], utilizes six switches, and its operation relies on series-parallel switching of capacitors. The one suggested in [9], has the ability to handle reactive power and exhibit a maximum utilization of DC bus voltage, which stands in contrast to HB based topologies. The topology uses only six switches and also has less TSV and THD (2.2%) than all other topologies presented in this category of TLI, making it more suitable for SPV applications. The topology presented in [10], also has a smaller number of passive components, but it has more leakage current than other topologies mentioned in Table 1. The topology presented in [11], ensures 100% utilization of DC bus voltage for the SPV system, but the TSV is around 6.5 volts. The topology suggested in [12], not only offers voltage-boosting capabilities but also includes a soft charging operation ability, making it appropriate for SPV-based grid- tied systems. The paper [14], proposed a family of four distinct inverter topologies. The important characteristic of this family is the achievement of 100% utilization of DC-bus, constant CMV and symmetrical operation. In paper [17], a simplified SC inverter is introduced, which utilizes a combination of a Marx inverter structure and an H-bridge. There is no need for bulky inductors, thus making the system compact. The topology introduced in [21], utilizes six switches and three SC to achieve only five-level output voltages. TSV (1 p.u.) and THD (3.7%) of this topology are better than presented in [15], [16], [17], [18], and [19]. Thus, this topology is also suitable for SPV applications. The topology presented in [24], utilizes four switches, making it the first five-level topology to use the minimum number of switches.

In paper [31], novel topology is derived by combining a 3-level front-end boost converter and a classical T-type inverter. The suggested topology mitigates high inrush as well as leakage current. The topologies mentioned in [32], [33], [34], [35], and [36] utilize the maximum number of passive components (more than 10), but the topology presented in [32] not only uses the maximum number of passive components but also

has the highest TSV among all the topologies as shown in Table 1. Thus, this topology is not a good choice for SPV applications. The topology introduced in [39], uses fewer components, and to address high charging currents during capacitor charging, it proposes the use of the QSC technique.

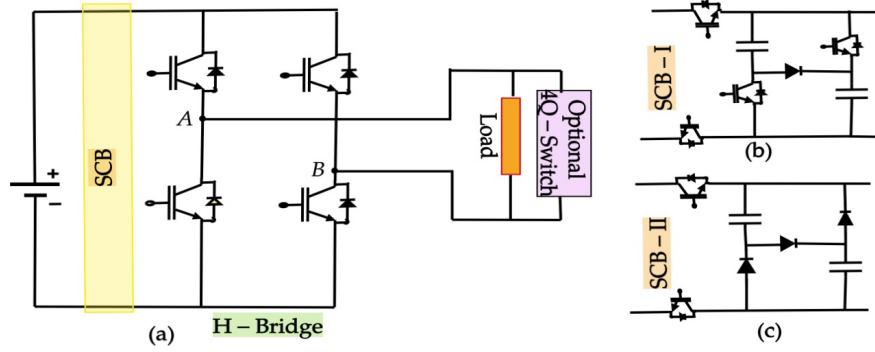


Figure 12: (a) Single-phase SC-5L-TLI topology (b) SCB-I (c) SCB-II.

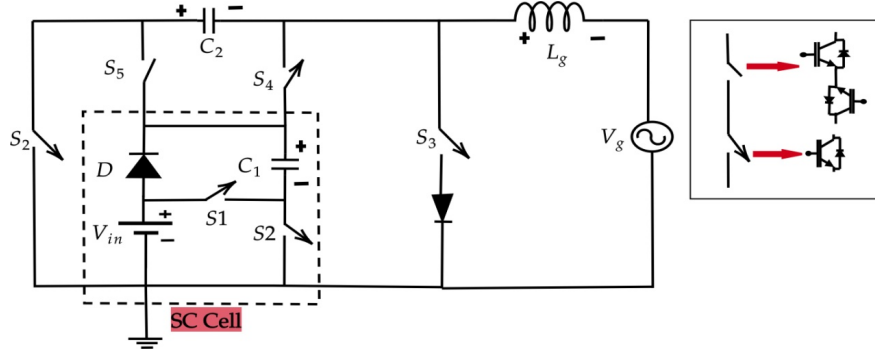


Figure 13: Five-level SC based TLI.

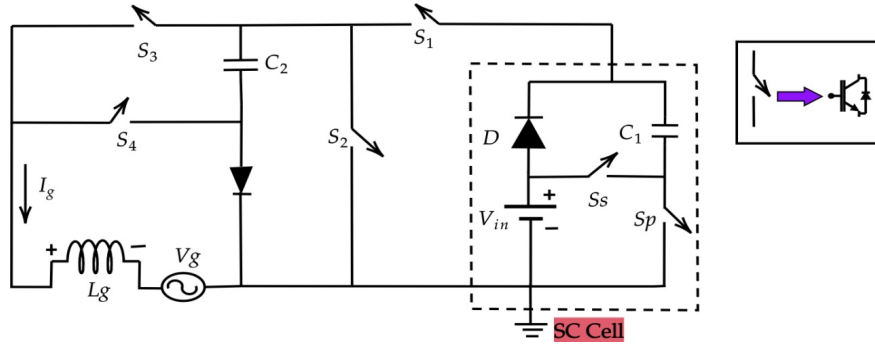


Figure 14: SC Based 5L-CG-TLI.

Table 1: Comparison of Common Ground Topologies

Ref.	Components	TSV	Ileakage	THD	Switching Scheme
S	D	C			

Ref.	Components	TSV	<i>I</i> _{leakage}	THD	Switching Scheme		
[6]	7	2	2	2V _{in}	< 17mA	-	PWM
[7]	9	0	2	2V _{in}	-	-	PWM
[8]	6	2	2	2V _{in}	0 mA	2.2 %	Unipolar
[9]	6	1	3	V _{in}	0 mA	3.5 %	SV PWM
[10]	6	2	2	-	8 mA	-	SLS PWM
[11]	7	0	3	6.5V _{dc}	0 mA	3.8%	Level Shift PWM
[12]	8	0	2	4V _{in} /6.5pu	0 mA	-	PWM
[13]	8	0	2	V _{in}	<100 mA	1.76%	Modified PWM
[14]	8	1	2	V _{in}	90 mA	2.3 %	Level Shift PWM
[15]	6	1	3	4V _{in} /6V _{dc}	0 mA	-	Unipolar PWM
[16]	6	2	3	4.5 p.u	0 mA	2%	PCC
[17]	7	0	1	-	0 mA	18.7 %	PWM
[18]	9	0	3	V _{in} /2V _{in}	0 mA	36.4 %	SPWM
[19]	7	2	4	V _{in} /2V _{in}	0 mA	2.92 %	Level Shift PWM
[20]	9	1	4	-	0 mA	-	SPWM
[21]	6	1	3	1 pu	0 mA	3.7%	PD-PWM
[30]	7	0	1	V _{dc} /2V _{dc}	0mA	1.03%	PDC PWM
[31]	10	0	2	8.5 p.u	0mA	-	PWM
[32]	8	1	2	-	0 mA	-	PWM
[33]	8	0	3	5.5 p.u	0mA	2.13%	SPWM
[34]	8	0	3	6.5p.u	0mA	1.1%	LS-PWM
[35]	10	0	2	2V _{in}	0 mA	2.5%	Level Shift PWM
[36]	6	3	3	V _{dc} /2V _{dc}	0mA	1.03%	PDC PWM
[37]	6	1	3	8.5 p.u	0mA	-	PWM
[38]	8	1	3	-	0 mA	-	PWM
[39]	6	1	2	5.5 p.u	0mA	2.13%	SPWM

S = Switch, D = Diode, C= Capacitor, L = inductor

4. H-Bridge-Type Topologies

4.1 Midpoint Clamped-Type Topologies

MPC-type MLI produce voltage levels from the neutral point voltage using clamping diodes. However, more active semiconductor switches, clamping diodes, and DC-link capacitors are needed as voltage levels increase. This results in high conduction loss as large number of clamping diodes are connected in series to block higher voltages and generation of reverse recovery currents, impacting switching losses of other devices. Additionally, NPC inverters with higher-levels face challenges with the balancing of DC-link voltage. Moreover, ANPC inverters divide the DC-link into two capacitors, thereby reducing the complexity of balancing DC-link capacitor voltages compared to conventional FC and NPC type inverters, which require four DC capacitors. The full bridge (FB) inverter is developed by incorporating semiconductor devices either at DC or AC side to achieve voltage clamping. These type of topologies are referred to as MPC TLI topologies. The main advantage of MPC technique is the reduced leakage current with lesser ripple compared to other topologies that have constant CMV. The MPC topologies like H5-D [47], OH5 [48] H5-HERIC [49] etc. are explained below:

The paper [47], presented an enhanced H5-D topology, which builds upon the existing H5 configuration by incorporating a clamping diode (D₁). This addition aims to address CMV fluctuations observed in original H5 topology and is depicted in figure 15. The switches S₂ and switch S₄ are activated during free-wheeling period. The voltage variation was limited to 0 V to 200 V despite not obtaining a single stable CMV. The

value of RMS leakage current is 13.88 mA. The efficiency is 97.3 %, surpassing that of the OH5. Out of all the zero state MPC topologies that were chosen, the THD performance was the best.

The paper [48], introduced OH5 topology as shown in Figure 16, where the two switches of the bridge particularly the upper ones, facilitate the free-wheeling phase of current flow. In this topology Switch S_6 is effectively utilized during free-wheeling phase that helps in maintaining constant CMV at 200 V. The suggested topology exhibits the lowest efficiency in its category as significant number of switches are operating at SF. The key benefit of proposed topology is the attainment of differential mode characteristic (DMC), which is analogous to the unipolar SPWM-FB grid-tied inverter, but with greater efficiency. Furthermore, adding more switches to DC side reduces the input voltage requirement by half. Thus, a constant CMV is obtained. The HERIC Active-I topology, introduced in [49], is illustrated in Figure 17. Switch S_7 is solely responsible for clamping the midpoint voltage in the zero state of the positive half cycle (PHC). The Switches S_6 and S_5 operate at 50 Hz frequency to establish an additional free-wheeling path utilizing the AC decoupling technique. The suggested topology has the maximum efficiency among the topologies chosen in this category because of smaller conduction losses. Leakage current has an RMS value of 41.41 mA (peak value 120 mA), which is higher than that of other selected topologies in this category. However, there is a relatively higher leakage current as only the PHC is clamped. The topology named HERIC Active-II (Figure 18) is proposed in [49]. Switch S_7 is used for clamping purposes and operates at the SF in the zero voltage state (ZVS). This topology demonstrates greater efficiency compared to PN-NPC and H5-D topologies.

The paper [50], introduced a MLI topology inspired by the concepts of HERIC and H5 topologies, with the primary objective of reducing the leakage current flow. The modulation technique designed for the suggested topology aims to eradicate the leakage current. The paper employs a modified H-bridge design incorporating two additional semiconductor components, S_6 and D_3 as illustrated in Figure 19. On the AC side, freewheeling path is established during the PHC using S_5 and D_4 . Conversely, on the DC side, a bidirectional switch is linked between the midpoint of the input capacitor and the midpoint of the leg which is formed by S_1 and S_2 .

The PN-NPC topology, introduced in [51], is illustrated in Figure 20. This configuration utilizes eight switches, with four operating SF and the other four at 50 Hz frequency. In both zero and active mode, the switches S_8 and S_7 are used for clamping purposes. This topology has the highest efficiency relative to OH5. The suggested topology can also inject reactive power, a significant advantage of future SPV inverters. It has a relatively low RMS leakage current of 13.29 mA.

Kerekes et al. [52], introduced an HB zero-voltage state rectifier (HB-ZVR) topology. This configuration (Figure 21) maintains CMV by integrating a ZVR into the conventional H-bridge inverter. The HBZVR-D topology, introduced in [53], is shown in Figure 22. The Clamping of the mid-point voltage is achieved using two diodes, D_5 and D_6 . During the zero state, switch S_5 creates the free-wheeling path which is facilitated by diodes D_1 – D_4 . The leakage current has an RMS value of 22.75 mA with peak value around 110 mA and the CMV remains approximately constant at 200 V, The Computed efficiency for this topology is 97.6 %, surpassing that of the H5-D and OH5 topologies.

This paper [54], introduced a novel HB clamped TLI designed for grid-connected SPV applications. The clamping circuit comprises two switches and a FB diode configuration, capable of clamping the AC terminal voltage to the DC midpoint during freewheeling. Consequently, the CMV remains constant, making it suitable for PV systems. The paper [55], paper suggested a re-oriented HB-ZVR-D topology (Figure 23) with reduced CMV and leakage current, resulting in improved performance and stability leakage current, enhancing overall efficiency, decreasing power loss, and ultimately reducing costs. This topology combines elements from both HB-ZVSCR [55] and HB-ZVR-D [54]. The proposed topology incorporates a MPC circuit consisting of a power switch and a FB diode, which stabilizes the CMV from zero to the DC midpoint during the freewheeling phase. Consequently, the CMV stabilizes, leading to decrease of leakage current.

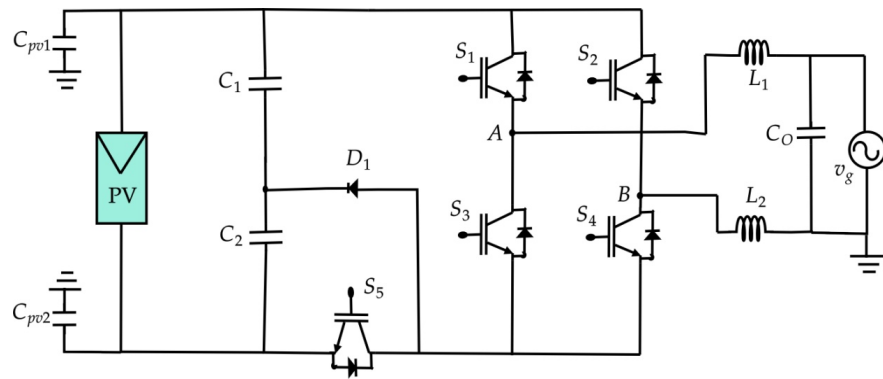


Figure 15: H5-D Topology.

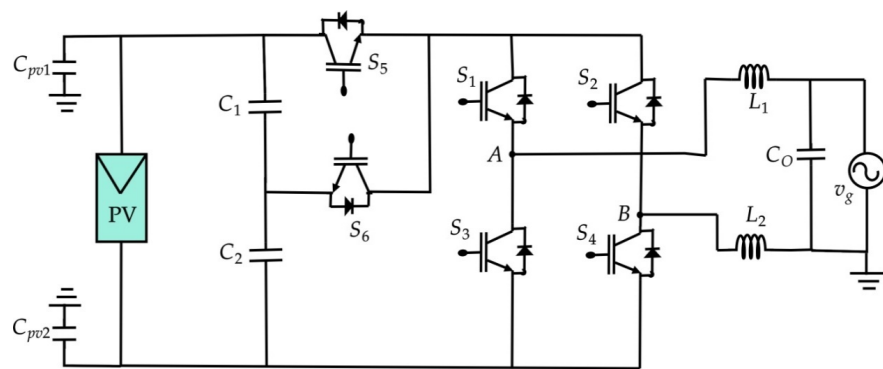


Figure 16: OH5 Topology.

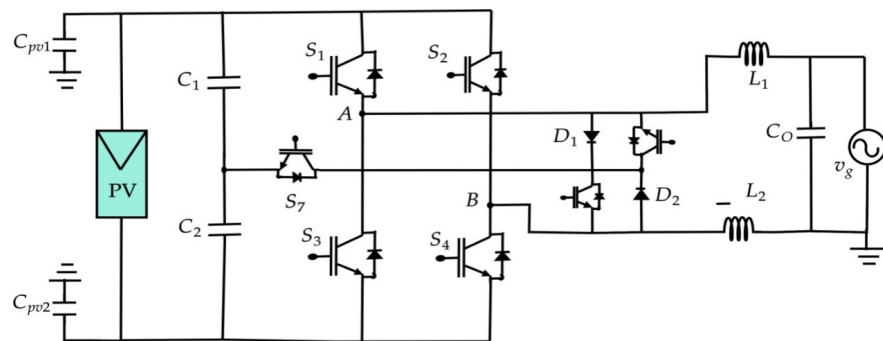


Figure 17: HERIC Active-I topology.

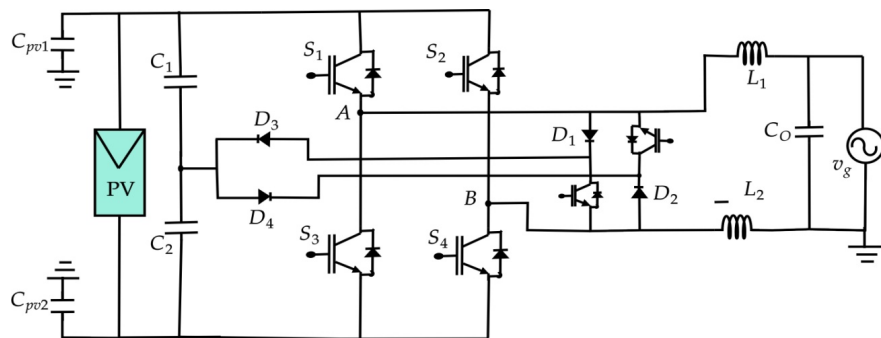


Figure 18: HERIC Active-II topology.

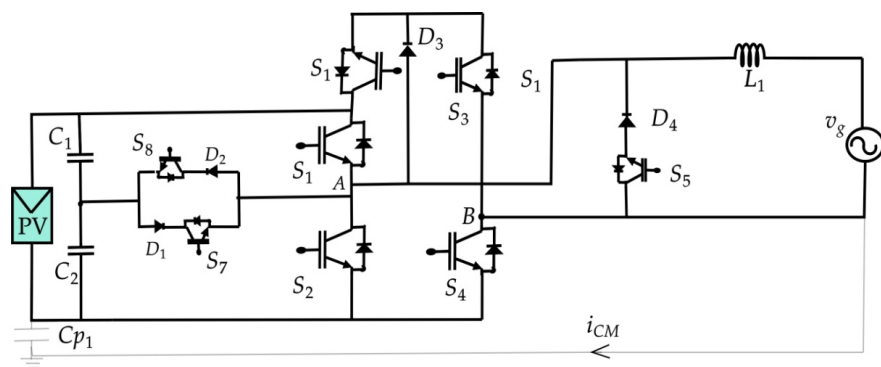


Figure 19: H5-HERIC topology.

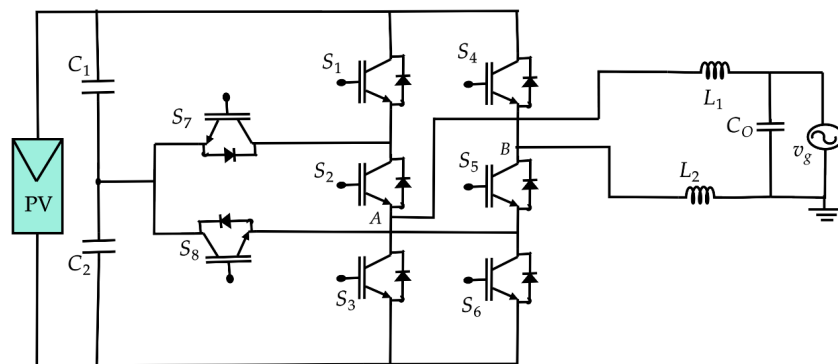


Figure 20: PN-NPC topology.

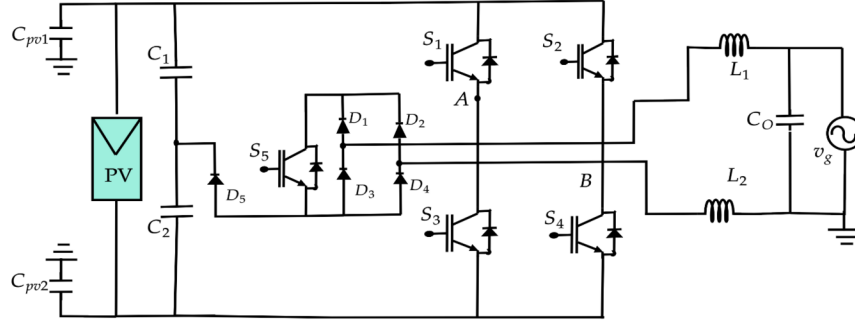


Figure 21: HB-ZVR topology.

In paper [56], authors introduced a double clamping branch on the input side, as depicted in Figure 24. The voltage can be clamped during the freewheeling phase by a capacitor divider on the input side. The novel topology introduced, eliminates varying CMV, requiring low-input voltage similar to bipolar PWM FB configurations. The suggested topology comprises of six switches and two diodes and can work with power factors in addition to unity.

The 5L-ANPC is widely recognized as one of the most popular topology among 5-level inverters. In paper [57], a novel 6-switch 5L-ANPC (6S-5L-ANPC) topology has been introduced. The suggested 6S-5L-ANPC inverter reduces the active switches by two compared to traditional 5L-ANPC inverters, by using only two diodes and six active switches. This lowers conduction losses. The reduction in the number of active switches compromises some reactive current paths. The suggested modulation scheme allows the 6S-5L-ANPC inverter to work effectively under both active and reactive power conditions. Additionally, the FC is optimized to accommodate both active and reactive power scenarios. This paper [57], highlights the benefits of the suggested topology by comparing it with conventional 5L-ANPC inverters across device efficiency, SF, voltage rating and system volume is crucial. The paper [58], presents a new 5-level boost-type inverter designed for wide range of applications such as pumps, rolling mills, tractions, mining, fans and notably grid-tied RE systems. This DC-link voltage requirement is halved compared to a conventional 5-level ANPC, NPC, and 5-level FC inverters. Additionally, it lowers the number of passive and active components while maintaining performance.

In paper [59], a hybrid-bridge TLI is developed by combining the HB module with a NPC module. This advanced configuration not only mitigates leakage current but also lowers the required bus voltage in comparison to conventional NPC or HB inverters. A SPWM technique without dead time is introduced to improve both system reliability and power quality.

In these types of topologies, the aim is to keep the CMV constant by linking the grid neutral with the middle of the DC link bus. This is done to lower the leakage current and keep it below the standard value [46]. The various MPC topologies are presented in Table 2 for easy comparison and analysis. The H5-D topology presented in [47] uses fewer switches than the others presented in this category. The THD and leakage current of this topology are below the standard value. The efficiency is greater than OH5 at 97.3%. The paper [48], introduced the OH5 topology, which offers a key advantage of achieving of a good DMC. Among HERIC-type topologies, HERIC Active-I [49] has the highest efficiency, HERIC Active-II has the highest leakage current of the other topologies presented in Table 2, and H5-HERIC [50] and PN-NPC [51] have relatively low leakage currents than HERIC Active-I and HERIC Active-II. The PN-NPC topology can inject reactive power, which is a major advantage of future SPV inverters. The paper [52], introduced an HB-ZVR topology. The efficiency for this topology is 97.6%, surpassing that of the H5-D (96%) and OH5 (97.3%) topologies. Among the HB-ZVR topologies presented in [52], [53], [54], and [55], the topology presented in [55], exhibits the lowest leakage current, as depicted in Table 2. The OH-1 topology presented in [56], has a lower leakage current and THD value and can work with different power factors. Thus, this

topology is more suitable for SPV applications.

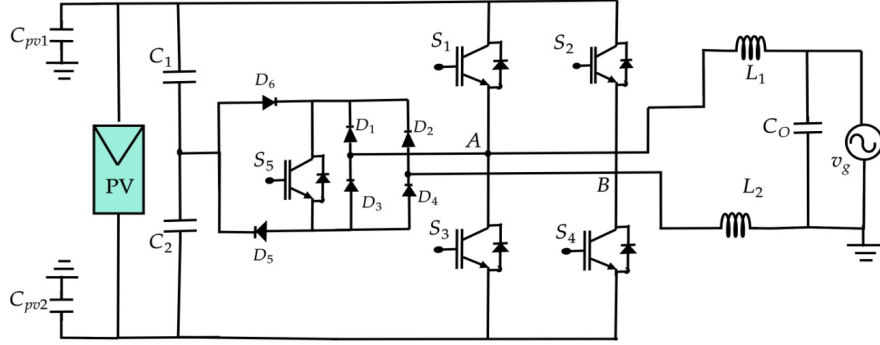


Figure 22: HBZVR-D topology.

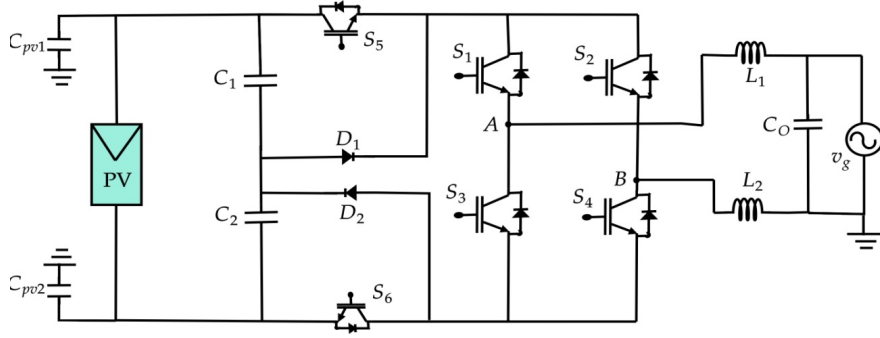


Figure 23: Re-Oriented HB- ZVR-D topology.

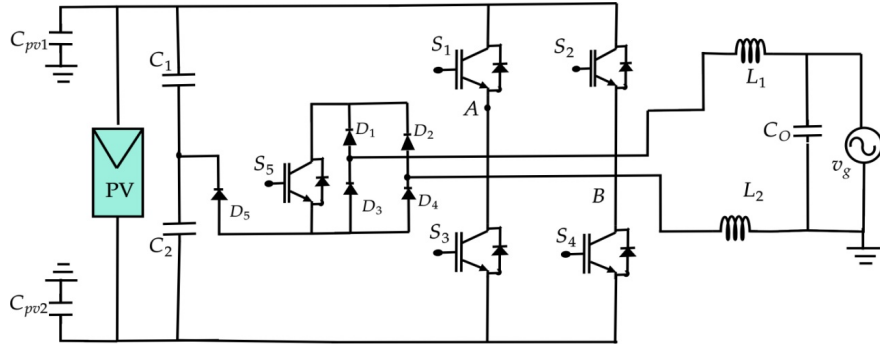


Figure 24: OH5-1 topology.

Table 2: Comparison of Midpoint Clamped Type Topologies

Ref.	Topology	Components	CMV	<i>I</i> _{leakage}	THD	Switching Scheme		
		S	D	C				
[47]	H5-D	5	1	2	(0–200) V	120 mA	4.88 %	SPW
[48]	OH5	6	0	2	200 V	65.05 mA	1.5 %	SV P
[49]	HERIC Active-I	7	2	2	(199 - 200) V	41.41 mA	-	PWM

Ref.	Topology	Components	CMV	Leakage	THD	Switching	Scheme	
[49]	HERIC Active-II	7	0	2	(100 - 200) V	71.48 mA	1.7 %	PWM
[50]	H5-HERIC	8	4	2	(0 - 600) V	<25 mA	-	SLS-I
[51]	PN-NPC	8	0	2	(199.3 - 201.1)V	13.29 mA	-	PWM
[52]	HBZVR	5	5	2	(163 to 200) V	27.8 mA	1.9 %	PWM
[53]	HBZVR-D	5	6	2	200 V	42.7 mA	1.9 %	PWM
[54]	HB-ZVSCR	6	4	2	(199 to 200) V	-	1.8 %	PWM
[55]	Re-Oriented HB- ZVR-D	5	6	2	(0 - 150) V	17.163 mA	1.97 %	PWM
[56]	OH5-1	6	2	2	(200 - 248) V	5.1 mA	1.1 %	PWM

4.2 AC Side Decoupling

These type of TLI topologies are expanded by incorporating additional switches and diodes on the AC side. In these configurations, the aim is to attain low THD in the output current and voltage. Furthermore, they lower leakage current by keeping CMV constant. Two prominent examples of such TLI topologies are the HERIC and HERIC AC-based topologies, as described in references [60], [68], and [72].

The HERIC topology, initially proposed in [60], is illustrated in Figure 25. The reactive power capacity was increased by incorporating the advanced modulation technique (AMT) from [61] and bidirectional clamping approach from [62]. Moreover, through the adoption of soft switching techniques from [63, 64] and enhanced modulation techniques from [65], the efficiency of this topology was improved. The two switches (S_5 , S_6) act as AC decoupling switches in both non-zero and zero states since they operate at 50 Hz frequency, whereas as the remaining switches run on SF. The topology is the most efficient decoupling topology because of the superior performance of S_5 and S_6 . The leakage current has an RMS value of 12.09 mA, while the peak value is approximately 125 mA. The CMV is also fluctuating. The voltage THD is greater than improved H6 and H5. The HERIC AC-based topology (Figure 26) resembles the HERIC topology, utilizing two diodes in series with switch S_5 and S_6 , as suggested in [66] and [67]. These diodes are employed to carry the output current during freewheeling period.

The H6-I topology was initially proposed in [68], as depicted in Figure 27. Advanced hybrid modulation techniques introduced in [69, 70] further enhanced its reactive power injection capacity. Additionally, a new control strategy [71] improved the behavior of the CMV. In this configuration, switches S_5 as well as S_6 operate in both free-wheeling mode and active mode, with diodes facilitating free-wheeling of current. The remaining switches are operated at the SF. This topology outperformed the improved H6 topology in terms of efficiency due to lower switching losses in switches S_5 and S_6 . The leakage current has an RMS value of 14.94 mA, with a peak value approximately equal to 150 mA. However, the THD performance was inferior than other selected zero state decoupled (ZSD) topologies except for H6-IV, depicted in Figure 28. The topology named H6-II was initially suggested in [68], as depicted in Figure 28. Further enhancement in reactive power capability was achieved using AMT. In this configuration, switches S_5 and S_6 are operated in both free-wheeling modes and active, akin to the Ho-I topology, with diodes facilitating free-wheeling of current. The remaining switches run at the SF. The key difference between H6-II and H6-I lies in the location of inverter output terminal. The leakage current behaviour and THD performance, however, were not as good as those of other chosen ZSD topologies.

A novel H6 topology based on IGBT known as H6-III was introduced by Islam and Mekhilef [74], designed to inject reactive power into the grid, unlike H6-II and H6-I, and is obtained from H4 topology and it incorporates two additional active switches (S_5 , S_6) as illustrated in Figure 29. The switch S_6 and S_5 provides a path for freewheeling current during the ZVS, ensuring the decoupling of the PV module from the grid. During positive grid voltage, switches S_1 and S_2 are switched diagonally at HF according to voltage polarity, while S_5 remain ON. During ZVS interval, switches S_1 , S_2 , S_3 , and S_4 are OFF. Hence, galvanic isolation is achieved between the grid and PV array. The CMV does not exhibit oscillation at zero crossing, remains constant and is equal to $V_{PV}/2$. Consequently, leakage current generation is nearly zero.

The paper [75], introduced a novel H6 topology comprising of six switches and two diodes, illustrated in Figure 30. The basic SPWM and double-frequency SPWM techniques were utilized to produce unipolar voltage and also ensures no leakage current. The switches (S_1 – S_6) operates at the grid or SF depending on the polarity of the grid voltage. In the presence of a positive grid voltage, S_1 remains continuously ON, while S_6 , S_3 , and S_2 remain continuously OFF. Conversely, S_5 and S_4 operate at the SF during the power transfer stage. Switch S_1 is only switch that remains ON during the freewheeling interval, and all other switches are OFF, hence isolation between grid and DC side is achieved and current during this period flows through S_1 and D_7 . The CMV fluctuates during freewheeling periods, indicating that leakage current is not mitigated completely.

In paper [76], a new modified CMV clamped topology named as H6-IV is presented. In this topology, diode D_3 and D_4 clamp the CMV to half of the DC link voltage during ZVS. The leakage current is completely mitigated as CMV stays constant throughout the operation of the inverter. In paper [77], the H6 in mid-switch topology is introduced, featuring four operational modes. It is illustrated in Figure 31. In this configuration, the switch S_6 is utilized to complete the circuit during the freewheeling phase. Switch S_6 and the body diode of S_5 establish the freewheeling path during the PHC. Conversely, in the negative half cycle (NHC), the body diode of S_6 is activated along with switch S_5 . The primary drawbacks of this topology include huge size of filter capacitor and the presence of high-ripple output waveform.

In AC decoupling topologies the leakage current is reduced by maintaining the CMV constant. The various AC decoupling topologies are presented in Table 3 for easy comparison and analysis. The HERIC topology [60], uses bidirectional clamping approach and AMT, thereby, increasing the reactive power capacity of the topology. The paper [68], introduced H6-I topology. The leakage current has an RMS value of 14.94 mA and the THD performance is inferior than other ZSD topologies except for H6-IV, as shown in Table 3. The paper [72], proposed H6-II topology, whose THD performance and leakage current is worse than other H6 topologies. H-III [74] and topology [77], utilizes less number of passive components as compared to others. The paper [74], presented a new H6 IGBT-based topology known as H6-III. The leakage current and THD is less for H6-III [74] and modified H6-IV [76] as shown in Table 3. In paper [77], new topology known as H6 in mid-switch is introduced, featuring four operational modes. However, the primary drawbacks include the large size of capacitive filter and the presence of high-ripple output waveform. Overall, considering all factors, HERIC emerges as the preferred topology in this group.

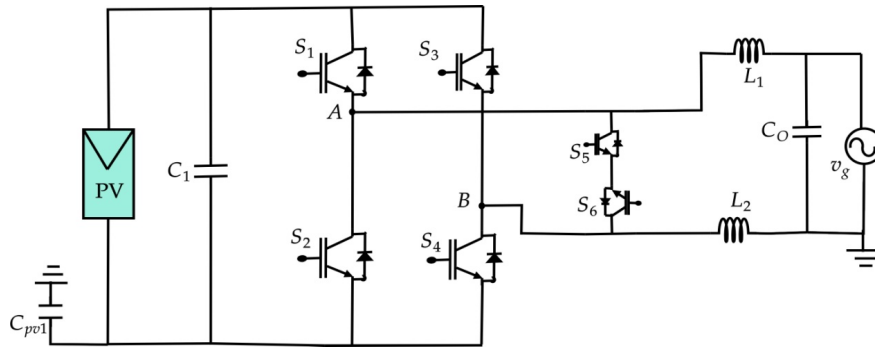


Figure 25: HERIC topology.

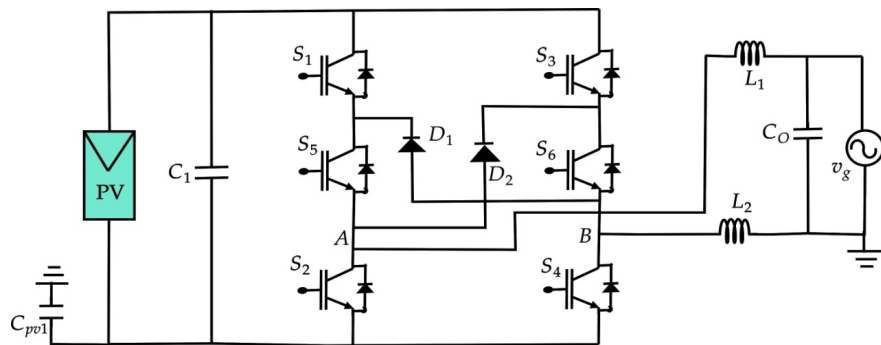


Figure 26: HERIC AC- Based topology.

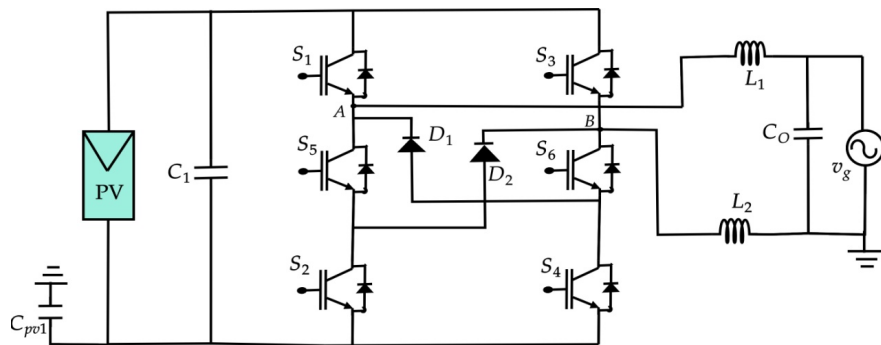


Figure 27: H6-I topology.

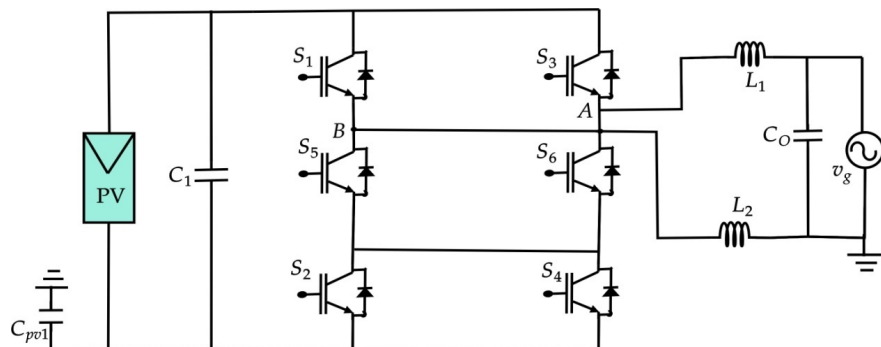


Figure 28: H6-II topology.

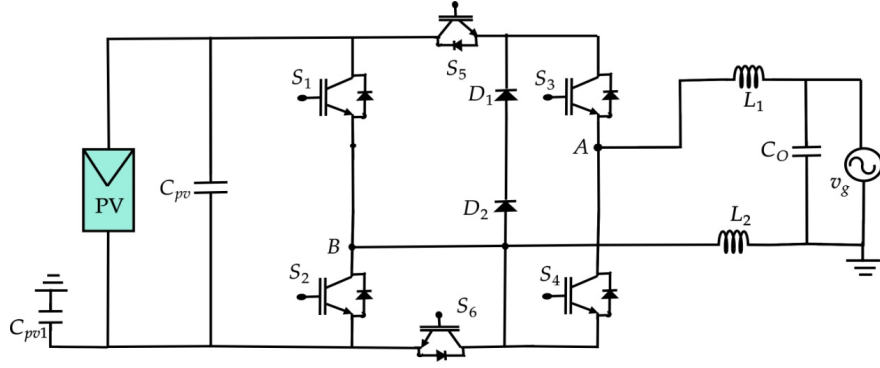


Figure 29: H6-III topology.

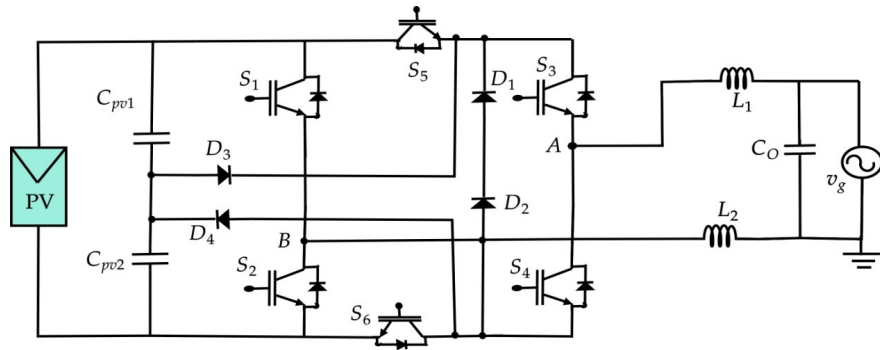


Figure 30: H6-IV topology.

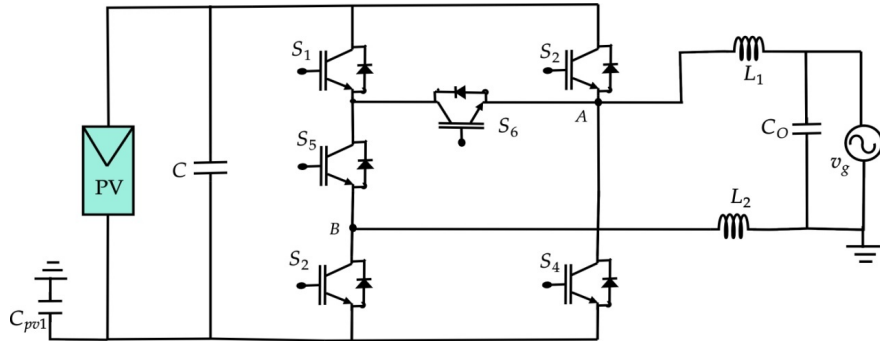


Figure 31: H6 In Mid Switch topology.

Table 3: Comparison of AC Side Decoupling Topologies

Ref.	Topology	Components	CMV	leakage	THD	Switching Scheme
		S	D	C		
[60]	HERIC	6	0	1	(165-235) V	12.09 mA
[66]	HERIC AC	6	2	1	(165-236)V	<200mA
[68]	H6-I	6	2	1	(159-240)V	14.94 mA
[72]	H6-II	6	2	1	(150-249)V	21.04 mA
						1.7 %
						-
						2.6 %
						-
						PWM
						PWM
						PWM
						PWM

Ref.	Topology	Components	CMV	<i>I</i> _{leakage}	THD	Switching Scheme		
[74]	H6-III	6	0	1	200V	9.8 mA	1.1%	PWM
[75]	H6-IV	6	2	1	150V	11 mA	2.8%	SPWM
[76]	Modified H6-IV	6	2	2	200V	6 mA	1%	PWM
[77]	H6 in mid-switch	6	0	1	(159-240)V	< 200mA	1.86%	PWM

4.3 DC Side Coupling

The incorporation of additional switches and diodes on the DC side contributes to the development of novel topologies, termed as DC decoupling-based TLI topologies. These configurations aims to reduce leakage current and achieve system balance. Several topologies fall under this category, including H5 topology, improved H6 topology and modified H6 topology.

The H5 topology was initially introduced in [78], is shown in Figure 32. The advancement in modulation techniques, as described in [61], and the adoption of bidirectional clamping, as outlined in [62], further improved the reactive power control of this topology. Additionally, strategies for reducing switching losses were implemented, such as soft switching approaches discussed in [79], along with the introduction of new modulation techniques outlined in [80]. In [81], optimization efforts were focused on filter settings and SF for the H5 topology design. Additionally, modifications were made to the FB topology structure, which included the addition of an extra switch S_5 . This addition isolated the grid during the ZVS from the PV side, effectively reducing leakage current. The upper two FB switches run at the 50 Hz frequency, while the lower two switches operate at the SF. The advantages of this topology are high efficiency, reduced RMS value of leakage current, and improved THD. However, the leakage current has RMS value of 13.23 mA, the peak value reached approximately 200 mA. Additionally, the CMV exhibited oscillations between 0 V and 400 V.

In paper [82], an improved H6 topology was introduced as shown in Figure 33. This modification of the H5 topology involved integrating extra switch on the negative side of the PV array to decrease leakage current relative to its predecessor. Operationally, the left bridge side functioned at line frequency, while the right bridge side operated at SF. The THD and leakage current is reduced relative to the H5 topology. However, the CMV fluctuated between 0 V and 400 V, with a peak leakage current of approximately 150 mA. Despite these improvements, the efficiency was slightly lesser than that of the H5 because of higher switch count.

In [83], a modified H6 topology was introduced, as illustrated in Figure 34, which features isolation of PV side from the grid during the free-wheeling phase. Operationally, the left side of the bridge functions at 50 Hz frequency, while the remaining switches operates at the SF. Notably, the CMV remains constant, resulting in minimal leakage current with values of rms = 12.05 mA and peak = 130 mA. It surpassed the performance of both the improved H6 topology and H5. The measured efficiency was greater than that of the improved H6 but equivalent to H5. Furthermore, it has higher THD compared to both improved H6 topology and the H5.

The DC decoupling topologies are presented in Table 4 for easy comparison and analysis. The three main topologies of this category are presented. These topologies possess high conduction losses and unbalanced switching, also there is need of additional devices to achieve the decoupling. The paper [78], proposed the H5 topology which offers high efficiency, improved THD and leakage current with lower RMS value, and as depicted in Table 4. The paper [82], introduced improved H6 topology where an extra switch is integrated on the negative side of the PV array. The efficiency was slightly lower than H5 due to higher switch count. The best topology for SPV application in this category is modified H6 presented in [83]. This topology possesses less leakage current and greater efficiency. The H5 topology possess less switch count relative to other topology.

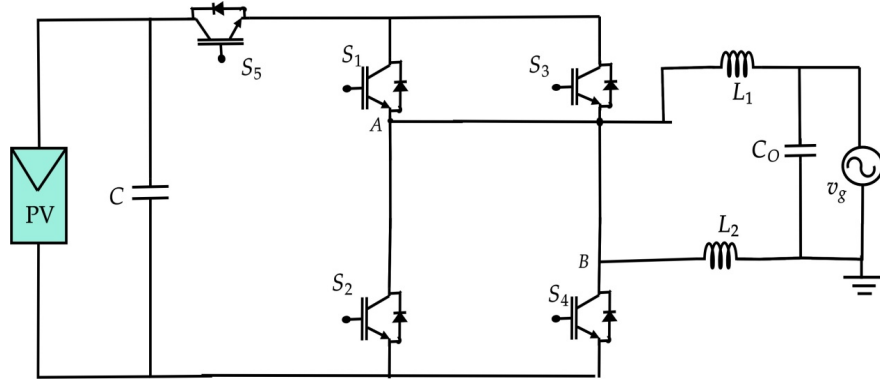


Figure 32: H5 topology.

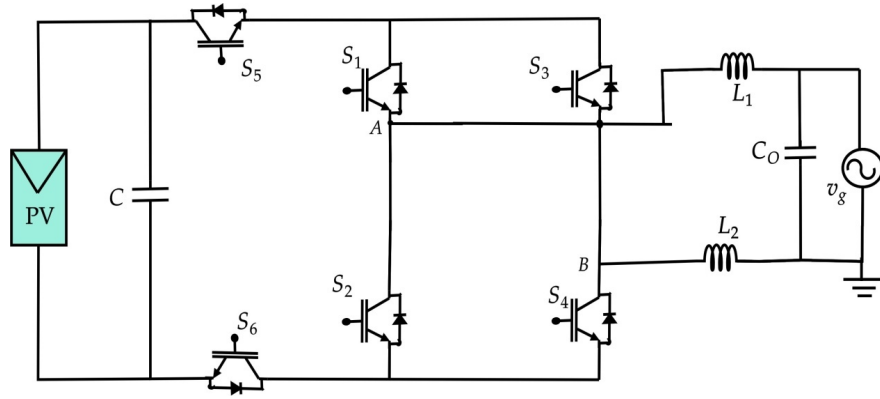


Figure 33: Improved H6 topology.

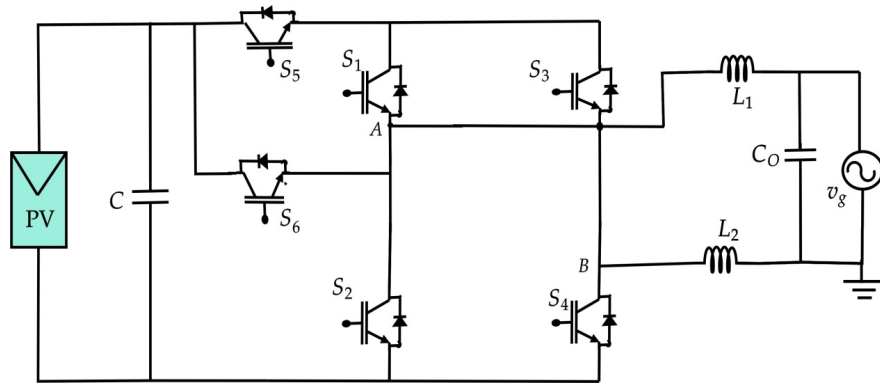


Figure 34: Modified H6 topology.

Table 4: Comparison of DC Side Coupling Topologies

Ref.	Topology	Components	CMV	<i>I</i> _{leakage}	THD	Switching Scheme
[78]	H5 topology	5	0	1	(0-400) V	13.23 mA
						1.8 % PWM

Ref.	Topology	Components	CMV	<i>I</i> _{leakage}	THD	Switching Scheme		
[82]	Improved H6 topology	6	0	1	(0-400) V	12.77mA	1.8 %	PWM
[83]	Modified H6 topology	6	0	1	(0-400) V	12.05mA	-	PWM

5. Buck–Boost Type Topologies

Buck-boost type (BBT) topologies are constructed by incorporating buck/boost and buck-boost (BB) configurations into the circuit, aiming to mitigate the complexity associated with TLI. In certain topologies such as presented in [86], there exists a direct connection between the PV panel negative terminal and the load, eliminating the flow of leakage current through the parasitic capacitor. However, it is noteworthy that these topologies may exhibit higher THD values.

The paper [87], presented a new CG TLI topology utilizing buck and boost operations from the DC-link. The configuration enables wide MPPT capability and enhances energy yield. The innovation lies in coupling a two-switch BB converter with a four-switch inverter, ensuring a constant voltage is maintained across the parasitic capacitances of the PV module. The connection is obtained without disrupting the operations of the DC-DC converter and inverter, resulting in very low earth leakage current. The paper [88], introduced a novel CG-type inverter that eliminates the need for SC. Instead, it employs an AC output capacitor and an inductor, both controlled by eight switches. By regulating the inductor with a sinusoidal duty cycle, the topology enables the generation of sinusoidal AC voltage in both modes.

The paper [89], introduced a novel single-phase TLI topology featuring a coupled inductor, tailored to fulfill the demands for greater efficiency, compact size, and cost-effectiveness. This novel topology addresses the need for an inverter with built-in step-up and step-down capabilities, making it appropriate for grid-connected SPV system. The key advantage include its straight forward symmetrical control scheme, common ground configuration, and excellent efficiency across a broad range of input voltages.

The paper [90], presented a new single-stage single- phase grid-connected BBTLI topology for SPV system. The topology incorporates a CG between the PV array and the grid neutral, effectively eliminating leakage currents. Additionally, it features buck-boost capability, facilitating maximum power point tracking even amidst wide variations in input voltage. Notably, the design utilizes inductor, enabling symmetric operation during both half cycles of the grid. Furthermore, two switches operate at 50 Hz frequency, reducing switching losses, while the remaining three switches function across various operational modes to minimize conduction losses.

In paper [91], the authors presented a novel TLI that has the ability to BB the input voltage. As illustrated in Figure 35, it contains four switches and two inductors. But in this design, DC offset current (DoC) is injected into the grid because one input inductor operates during the PHC and the other during the NHC. Furthermore, in this architecture, the THD (>5%) is higher than the IEEE standard.

A TLI based on the BB concept is presented in paper [92], with five switches and a single inductor. But it needs three more diodes, which means there are significant power losses, resulting in a less efficient system. The paper [93], suggested a doubly grounded (DG) inverter topology that can function with a wide voltage range of SPV source. But it needs eight controlled switches and an input inductor. Since, adding more switches to a topology usually results in lower efficiency and reliability as well as higher overall expenses.

A different BB based inverter design with five switches is suggested in article [94], to support a wider range of operation for SPV system. However, the suggested topology does not support double grounding of system. Moreover, three switches will be conducting in each mode, increasing the system's losses. This topology also requires a high rating inductor, which makes the system bulkier and reduces efficiency.

A DG-BB inverter topology that utilize only four switches is presented in article [95]. However, asymmetrical operation of this topology for negative and positive grid cycles is one of its key drawbacks. The inverter topology based on coupled inductor that can provide a significant boost in the output voltage is introduced

in article [96]. But the topology has a major disadvantage that it prevents the system from being doubly grounded. Furthermore, all three switches will be active during each switching interval, resulting in more power losses and less system efficiency. In addition, it shows asymmetric inverter operation, which causes a DC offset in the grid current. In order to lower leakage current and DoC, authors presented a novel DG-TLI topology in paper [97]. This design uses six switches, an inductor and a capacitor. However, the system becomes bigger, more expensive, and less efficient because of higher count of power components.

A new TLI structure is presented in article [98], which provides buck-boost voltage capabilities with just five switches. Although this design enables the system to be doubly grounded, it requires the installation of two more diodes, two capacitors and two inductors. Additionally, it adds DoC to the grid. A novel Cuk converter-based TLI topology with the double grounding characteristic is introduced in paper [99]. It makes use of five power switches, two additional diodes, an auxiliary capacitor and an input inductor. The suggested topology does not support reactive power. In the paper [100], a four-switch TLI topology with BB voltage capabilities is suggested, which uses three capacitors and two more inductors, making the system less reliable and more expensive. It also introduces a DoC into the grid and is devoid of the double grounding characteristic.

To reduce leakage currents, a DG five-switch Cuk converter-based TLI topology is presented in paper [101]. However, this topology is similarly affected by injected DoC into the grid. Furthermore, one auxiliary capacitor and two additional inductors are required, increasing the component count. The single-stage, DG-BB TLI presented in papers [102]–[103] requires high number of switches and additional power components (diodes, capacitors and inductors) to track the maximum power point (MPP) during wide variations in weather. These additions raise the size and cost of system and also power losses. The authors of papers [104]–[108] presented novel TLI configurations in an effort to optimize the amount of power devices and components. But because these TLI topologies use separate inductors for PHC and NHC, they suffer from the asymmetry of inverter operation, which introduces a DoC into the grid. To overcome this issue, paper [109]–[113] introduces impedance source-based TLI with double grounded features. However, these inverter topologies need numerous power components like switches, diodes, capacitors and inductors to BB the output voltage and track the MPP across a wide range of weather scenarios. In the paper [114], a new single-stage DG-TLI topology capable of BB of output voltage is proposed. This configuration successfully suppresses leakage currents since the neutral of grid is directly linked to the neutral terminal of PV array. Additionally, maximum power point tracking throughout a wide range of weather situations is made possible by the BB voltage capability of inverter. Additionally, the suggested inverter design operates symmetrically in both half cycles, adding a very little amount of DoC to the grid. This symmetrical operation obtained by using a single inductor for both half cycles of the grid. The suggested topology is implemented with five switches, two of which run at 50 Hz frequency and the remaining three at HF, resulting in lower overall losses.

This paper [115], presented a novel Boost type CG inverter that incorporated SC and quasi-Z-source (qZS) techniques. The topology uses a front-side qZS unit to provide a stable charging voltage and restrict the charging current to an SC inverting unit. The suggested inverter can function in both boost and buck modes by choosing appropriate duty ratio and modulation index values. In this topology, the capacitors within the SC unit are charged via the inductors of the front-side qZS unit, substantially reducing the inrush charging current that normally occurs in conventional SC-based CG inverters. The topology is suitable for low-input voltage sources as it has enhanced voltage gain

These topologies are suitable when there is a need to incorporate buck\boost function in a single circuit. This helps to eliminate the conversion stage required prior to the inverter. In most of the topologies there is an introduction of DC offset in the grid current. The THD as well as the component count is high as compared to other type of TLI topologies. The Buck Boost topologies are presented in Table 5 for easy comparison and analysis. The topology presented in paper [92], has high power losses and consequently, a less efficient system due to additional diodes. The topology suggested in [93], is capable of operating with a wide voltage variation of SPV source, but requires more number of switches. The one suggested in paper [94], does not allow system to be the DG and has high losses. The research paper [96], presented a novel

topology that not only possess higher efficiency but also utilizes less switches and other passive components. But one major drawback of this topology is that it prevents the system from being doubly grounded. In paper [99], a Cuk converter-based TLI topology with double grounding feature is introduced and it does not support reactive power. A new configurations of TLI are introduced in papers [104]–[108]. However, these TLI topologies suffer from the asymmetry of inverter operation, thereby introducing a DoC into the grid. To address this drawback, impedance source-based TLI with DG feature is presented in papers [109]–[113]. The highly efficient topologies in this category are [95–96], [101], [103], [107], [108] and [114]. The efficiency of these topologies is greater than 94% as shown in Table 5. Thus these topologies are suitable for SPV application. The topology presented in [114] (shown in Figure 36) is ideal choice for SPV application as it not only provides boosting but also reduces leakage current. It also injects negligible leakage DoC into the grid as compared to others. The passive components used in this topology is also less as mentioned in Table 5.

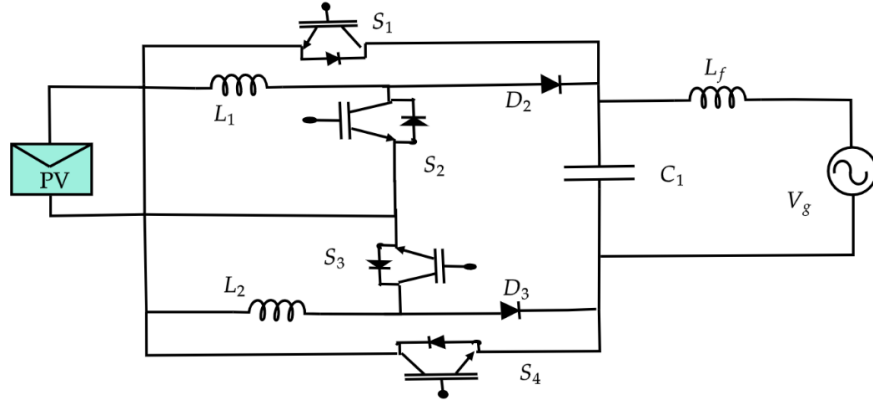


Figure 35: TLI topology with buck-boost voltage capability.

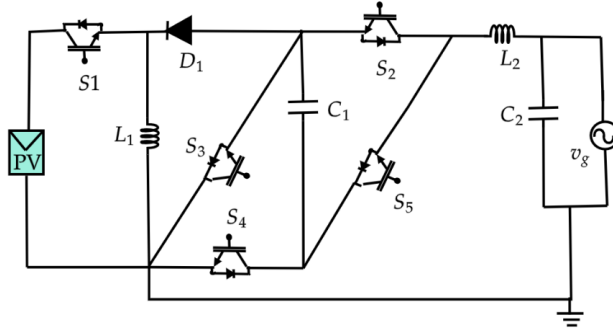


Figure 36: A single stage DG-TLI topology.

Table 5: Comparison of Buck-Boost Type Topologies

Ref.	Components	Double Grounding	THD	Reactive Power Capability	Efficiency
	S	D	C	L	
[91]	4	2	0	2	No 6.5% No 83%
[92]	5	3	0	1	Yes 4.9% No 80%
[93]	8	0	0	1	No - Yes 91%

Ref.	Components	Double Grounding	THD	Reactive Power Capability	Efficiency			
[94]	5	2	0	1	No	6.1%	No	88%
[95]	4	0	1	1	Yes	3.3%	Yes	94%
[96]	3	5	5	3	Yes	3.5%	Yes	94%
[97]	6	0	1	1	Yes	4.3%	Yes	90%
[98]	5	2	0	2	No	4.1 %	No	89%
[99]	5	2	1	2	Yes	3.43%	No	93%
[100]	4	0	1	2	Yes	3 %	Yes	94.5%
[101]	5	0	1	1	Yes	3.5%	Yes	93%
[102]	6	2	3	2	Yes	2.9%	Yes	94%
[103]	5	3	1	2	Yes	3.92%	Yes	93.5%
[104]	4	2	3	1	Yes	4%	No	91%
[105]	6	2	2	2	Yes	2.5%	Yes	94%
[106]	3	3	3	1	Yes	2.35%	No	93.5%
[107]	4	0	2	1	Yes	2.1%	Yes	94%
[108]	6	0	1	1	No	-	Yes	95.5%
[109]	4	0	4	2	Yes	-	Yes	93%
[110]	6	3	1	2	No	3.1%	Yes	91%
[111]	4	3	3	5	No	0.7%	Yes	92.8%
[112]	3	4	6	3	Yes	-	Yes	90%
[113]	4	3	3	4	No	-	Yes	85%
[114]	5	1	1	1	Yes	2.18%	Yes	94%

Topology	Advantage
CG	Absence of CMV effect Less semiconductor devices are used Small filter needed
MPC	CMV is constant minimum leakage current
AC – Decoupling.	conduction losses are less Output current does not flow through anti-parallel diode in FB THD is low
DC – Decoupling	DC bypass switch disconnects PV panel during leakage current
Buck- Boost	Low leakage current

6. Comparison of Various Transformerless Inverter Topologies

In the earlier sections, all types of TLI topologies were discussed in detail. In this section, a general comparison of all these TLI topologies on the basis of advantage, disadvantage, reactive power capability, inverter size, and efficiency is done as shown in Table 6. The CG topologies and AC decoupling topologies has high efficiency but these is complete elimination of leakage current in CG topology. Also CG topology use less number of semi-conductor devices, making is less costly, small in size and light weighted. This make CG topology highly recommendable for SPV applications. Table 6: Comparison of Transformerless Inverter Topologies

7. Authors Inference and Future Recommendations

TLI topologies are classified into three groups within this paper. To ensure a fair comparison between the topologies, tables were generated based on different parameters like component count, leakage current, THD, TSV, CMV of each topology. All CG topologies ensure zero leakage current. However, their topological design is most complex, requiring a direct connection between the PV system and grid. The CMV of this

category is typically zero. The CG topologies excel in mitigating the leakage current issue, yet they feature more intricate circuit configurations and control systems compared to other groups. The recommended topologies in this category are [9], [11], [13], [17], [24], [36]. In zero state MPC topologies, the peak leakage current is relatively smaller compared to ZSD topologies, albeit with a more intricate arrangement. OH5 and HBZVR-D topologies ensure constant CMV, while others in this group show partially constant CMV. Despite a higher mean RMS leakage current compared to ZSD topologies, the mean peak leakage current is reduced because of MPC technique. Among these, HERIC Active-I demonstrates the highest efficiency, while PN-NPC and OH-1 exhibits the best leakage current performance. However, when considering all factors, PN-NPC and OH-1 can be regarded to be the best topology in this group. In the decoupled topologies category, the absence of a clamping branch simplifies the structure of circuit. The CMV for all topologies in this category swing between 0 V and 400 V. The most efficient topologies in this group are those based on HERIC. The RMS value of leakage currents ranges from 12 to 15 mA. Among these topologies, modified H6 exhibits the best distribution of losses across switches. Overall, considering all factors, HERIC emerges as the preferred topology in this group, although H5 also performs well. TLI technologies have emerged as promising solutions for various renewable energy and power electronics applications because of their compact size and greater efficiency. Furthermore, there are still areas for improvement and future directions that can enhance their performance, reliability, and applicability.

- One key area of focus is the mitigation of leakage current, which remains a major challenge in transformerless inverters. Future research should aim to develop more effective techniques to reduce leakage current to negligible level, thereby improving safety and grid compatibility.
- Simplification of topologies is also a worthwhile pursuit. By reducing the complexity of transformerless inverter circuits while maintaining performance, reliability, and cost-effectiveness, future designs can be more accessible and easier to implement. The SC based inverters also can develop the multilevel output.
- Enhancement is another important aspect that needs attention. Although TLI are known for their high efficiency, further optimization can be achieved through advancements in semiconductor devices, control strategies, and thermal management techniques.
- Grid compatibility is crucial for transformerless inverters, especially in applications like grid-connected SPV systems. Future developments should focus on ensuring seamless integration with the grid, addressing issues such as synchronization, power quality, and compliance with standards and regulations.

In summary, continued research and innovation in transformerless inverter technologies hold significant potential for advancing renewable energy integration and power electronics applications. By addressing key challenges and pursuing future recommendations, transformerless inverters can play a vital role in renewable energy system in future

8. Conclusion

The development in TLI are vibrant due to their compact size and greater efficiency. However, leakage current remains a significant issue associated with TLI. Various techniques have been introduced to mitigate leakage current, but selecting the right technique for a particular inverter structure poses a challenge. This paper aims to analyze the causes of leakage current generation and its variation under different scenarios in detail. This paper provides a comparative analysis of recent TLI topologies for grid-connected SPV system. The topologies are categorized based on methods for reducing leakage current. Performance metrics such as CMV, leakage current, THD and efficiency are used for comparison. The key findings of the study are as follows:

- The CMV of CG topology is typically zero and hence excel in mitigating the leakage current issue.
- Topologies that offer solely galvanic isolation are incapable of entirely eliminating leakage current.
- CMV-clamped topologies, such as, passive-clamped H6, optimized H5 and active-clamped HERIC, effectively mitigate leakage current by providing both CMV clamping and galvanic isolation.

- DC-side decoupled topologies are less efficient than AC-side decoupled topologies.
- All topologies generate sinusoidal current which is synchronized with grid voltage.

This paper will serve as a valuable resource for new researchers, providing a comprehensive understanding of existing topologies, common-mode leakage current reduction techniques, and avenues for developing new topologies.

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